

MagPixel: Modular Toolkit for Designing Interactive Magnetic Shape Displays

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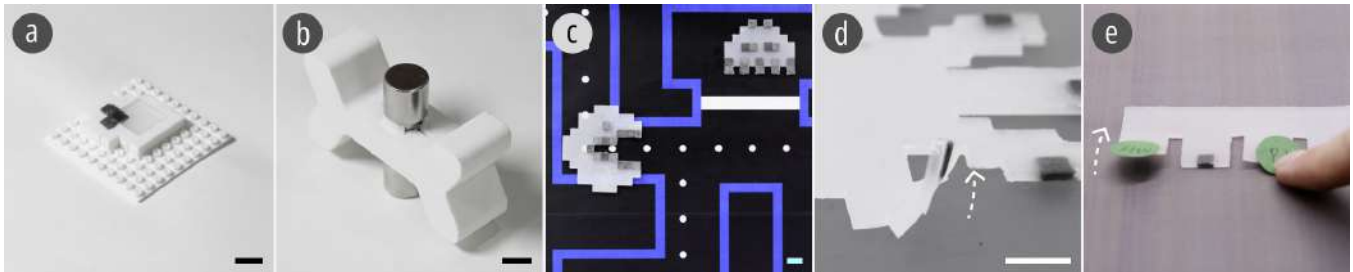


Figure 1: (a) Fabrication of a *MagPixel* pattern using our toolkit. (b) Magnetization of a magnetic pixel using our magnetizer. (c) A tangible version of the video game Pac-Man using the toolkit from *MagPixel*. (d) A close-up of the clock application. (e) The interactive reminder application. (Scale bar: 10 mm.)

ABSTRACT

This paper introduces *MagPixel*, a modular toolkit for democratizing the design and fabrication of interactive shape displays using magnetic soft materials in the realm of Human-Computer Interaction (HCI). Known for their instant responsiveness and flexibility, magnetic soft materials have been used in material science and soft robotics but remain under-explored in HCI. *MagPixel* addresses

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this gap by offering a user-friendly, cost-effective solution for creating morphing tangible interfaces actuated by external magnetic fields. Existing methods of fabricating magnetic soft interfaces are complex, requiring multiple specialized tools and skills for casting, magnetization, and assembly. By simplifying fabrication into straightforward, standardized procedures, *MagPixel* eliminates the need for proprietary materials and extensive resources. The toolkit provides DIY enthusiasts in the maker community with the means to effortlessly create and modify interactive magnetic soft displays, *MagPixel* paves the way for broader experimentation and innovation in interactive design and HCI.

CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

KEYWORDS

Shape-changing interfaces, magnetic materials, tangible interfaces, fabrication

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1 INTRODUCTION

Magnetic soft materials have become a popular topic in the fields of materials science and soft robotics for their unique properties. They are instantly responsive due to the nature of magnetism and are flexible for their elastic base materials, which hold the potential for integrating interactive shape display and tangible interaction. In human-computer interaction (HCI), some studies have utilized this type of materials to create passive haptic patches [8] and self-aligned paper [11]. However, the potential of magnetic soft materials for expressive information display, arising from their flexibility and shape-changing ability, is still an under-explored area. Additionally, making magnetic soft materials is challenging for personal fabrication due to their intricate procedures, meticulous design, and complicated tools.

Therefore, we present *MagPixel*, a modular kit for fabricating interactive shape displays composed of magnetic soft pixels. By providing a low-cost, easy-to-use toolkit, *MagPixel* enables users to create a morphing tangible interface activated by the external magnetic field. The material of 100 magnetic pixels costs approximately 1 USD, and the toolkit costs around 5.5 USD.

Previously, in material science and robotics, fabricating magnetic soft materials is challenging due to complex mold-making, casting, magnetization, and assembly, each requiring separate tools and specialized training or experiences. In contrast, with the toolkit brought by *MagPixel*, the fabrication pipeline can be streamlined into easy-to-follow and standardized procedures. Meanwhile, our method is accessible because it does not require proprietary material that is difficult to obtain; it is also affordable, necessitating only the minimum of tools and resources necessary for making magnetic pixels. Accompanied by the fabrication method is the design process, which is based on a two-dimensional grid system that describes the pattern of the shape display. The modular design enables flexibility and scalability, which invites customization through a modular assembly method. The tile-based toolkit can also avoid ambiguity and produce consistent results, and it is easy to modify or duplicate the design based on the user's needs.

This paper's contributions include:

- (1) A grid-based design and fabrication pipeline and easy-to-use modular toolkit for making shape displays and tangible interaction using accessible magnetic soft materials, promoting the ease of customization through straightforward handicrafts.
- (2) Design primitives of *MagPixel* patterns for reference, capable of achieving various reproducible shape-changing effects.

- (3) A series of implementations that showcase the interactive shape displays enabled by *MagPixel*.

2 BACKGROUND AND RELATED WORK

In HCI, magnets are powerful tools for creating interactive things, and have been used for easy-to-construct prototyping [2, 5, 6], creating haptic interfaces [10, 12, 13, 17, 23], and connectable circuits [1]. Recent works have also utilized magnetic materials that can be programmed for haptic interaction [8, 9, 19, 20], graphic display [9, 20], self-alignment/assembly [9, 11], and pin-based shape display [21]. However, existing interfaces and interactive objects are relatively rigid, resulting in limited morphing abilities; the majority of the studies have yet to utilize a truly flexible material; most used magnetic sheets that are rigid compared to materials like silicone. In contrast, our work explores untethered, shape-changing structures made from magnetic soft materials, which can be used for three-dimensional shape displays and tangible interaction.

Researchers have laid the foundation for making magnetic soft materials with cutting-edge technologies. In material science and soft robotics, studies have utilized composites of magnetic particles (neodymium-iron-boron - NdFeB, Fe₃O₄ or Carbonyl Iron, etc.) and shape memory polymers, elastomers, or other soft base materials to fabricate morphing structures controlled by an external magnetic field; the fabrication methods include molding [7], ink writing [3, 14], ultraviolet (UV) lithography [18], and laser cutting [16]. Previous works have presented various actuators and soft robotics designs made from these materials and their applications, especially in the biomedical domain [4, 15].

However, to fully explore these materials within the scope for HCI, we have to solve these problems: 1) Existing fabrication methods involve specialized materials that are difficult to obtain and require expertise, and expensive devices such as an impulse magnetizer, which typically costs thousands of US dollars or more. 2) Existing designs of soft robotics or other types of morphing objects require domain knowledge of modeling and simulation, making designing tangible shape displays out of reach for average users. We address these problems by providing a low-cost, easy-to-use fabrication toolkit and corresponding modular design method to enable novice makers to get reproducible results.

3 DESIGN AND FABRICATION METHODS

3.1 Modular Molds

We provide a modular mold toolkit (Figure 2) for fabricating magnetic pixels, which will be assembled into magnetic soft interfaces. This modular mold toolkit contains three 3D-printed parts: the wall units, the floor units, and the baseboard. As shown in Figure 2, the wall unit (Figure 2a1) and the floor unit (Figure 2a2) are polyomino blocks with different heights, 6.50 mm and 5.00 mm respectively. The wall units and floor units can be pinned down on the baseboard (Figure 2b) to hold an uncured silicone mixture with a shape that matches all the connected floor units. The pixel unit produced by the mold toolkit is sized in $5 \times 5 \times 1.5$ mm. This modular feature allows users to design magnetic soft patterns through natural and intuitive hands-on assembly without needing a specific pattern design in the first place. The methods of assembling molds and design primitives will be presented later in Section 3.2 and Section 3.3. The

3D printing process of this study is performed using Raise3D E2 as the printer and Polymaker PolyTerra PLA.

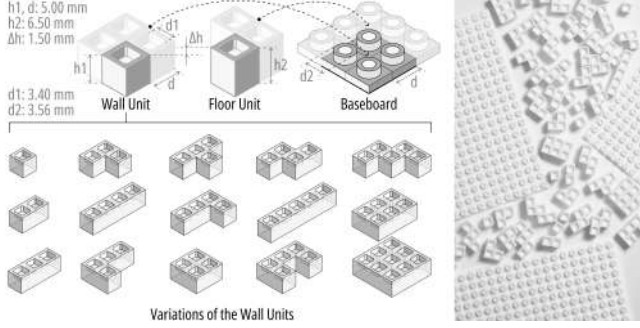


Figure 2: Modular mold toolkit of *MagPixel* contains the wall units, the floor units, and the baseboard. Variations of the wall units are demonstrated.

3.2 Fabrication of *MagPixel*

To fabricate magnetic pixels, we use NdFeB powder (maximum particle size is less than 200 μm , 66.7 wt%) and silicone rubber (Ecoflex 00-30, Smooth-on, Inc). We begin with mixing part A and part B of silicone in a 1 to 1 mass ratio to create a silicone mixture. This mixture will also be used to connect the magnetic pixels. Then, we thoroughly combine and stir the NdFeB powder with a portion of the uncured silicone mixture in a 2:1 mass ratio until we acquire a viscous mixture. Achieving an even distribution of the NdFeB particles in the mixture is important for a consistent shape-changing result. This mixture is then slowly poured into the mold to cast a single pixel and heated in an oven at 50 $^{\circ}\text{C}$ for 20 min. The final results are individual pixels with a dark grey color, sized in 5 $\times$ 5 $\times$ 1.5 mm.

We started by investigating the relations between NdFeB powder-silicone mass ratios and the magnetic strengths on casted magnetic pixels. As the mass ratio increases, the magnetic strengths increase, while the flexibility of the pixels decreases. Considering these factors and the cost of the materials, we chose a 2:1 mass ratio. In addition, heating at 50 $^{\circ}\text{C}$ for 20 min can solidify the mixture sufficiently without deforming the thermoplastic PLA mold toolkit.

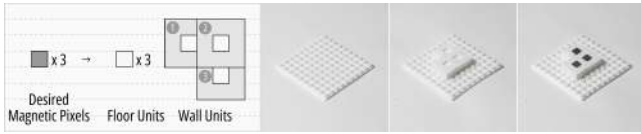


Figure 3: Fabrication of magnetic pixels. Three floor units and the surrounding wall units are installed on the baseboard to form three pockets, which will be used to cast three individual pixels.

We offer a magnetizer to quickly magnetize the pixels, using 3D printed modules and NdFeB magnets (N52, $\varnothing 15\text{ mm}$) that are commonly available. As illustrated in Figure 4, the magnetizer contains two identical 3D printed magnet holders, with a square void

on the bottom sides that matches the dimension of a pixel square. For each magnet holder, three N52 magnets (two magnets with $H = 5\text{ mm}$, and one with $H = 20\text{ mm}$) are installed to reach the magnetic field strength of 790 mT, which we tested at the square void after the magnetizer is connected. Before initiating magnetization, the magnetic pixel holder is placed within the void on the underside of the magnetizer, aligning its center with the two magnets. The magnetization process will be finished after the two parts of the magnetizer are coupled for approximately 2 seconds, and the magnetized magnetic pixel can be retrieved after the magnetizer is separated. This should be proceeded at room temperature to prevent heat-induced demagnetization. Repeat the process of magnetizing until the desired number of magnetic pixels is acquired. The magnetic pixel can be magnetized to reach a magnetic strength of 25 mT to 28 mT.

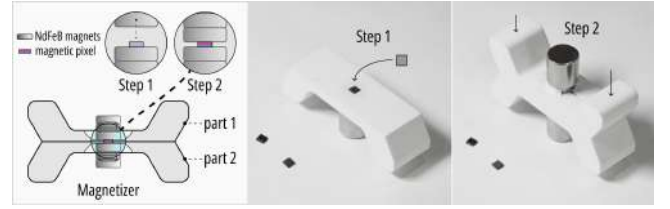


Figure 4: Magnetization. After curing, the three magnetic pixels are inserted, one at a time, into the square void of the magnetizer. Connect the two parts of the magnetizer to finish the magnetization process. This figure the fabrication process of a shape-display pattern.

The main design constraint is the XY dimension of the pattern, which has to fit the build platform of the selected 3D printer. After the magnetized magnetic pixels are placed at the target position within the mold, the next step is to pour an adequate amount of thoroughly mixed silicone (Ecoflex part A and Ecoflex part B in a 1:1 ratio) into the mold until filled. Finally, a complete shape display is successfully created after the second curing process at 50 $^{\circ}\text{C}$ for another 20 min.



Figure 5: Fabrication of the *MagPixel* pattern. Assemble the floor units into the desired pattern on the baseboard, then install the wall units to make the mold containable. Finally, add the uncured silicone until the magnetic pixels are submerged and the mold is filled.

3.3 Exploring the Grid-Based Design Method

Using the modular toolkit *MagPixel*, we explored the grid-based design method by fabricating primitives composed of single magnetic pixel, duo, and trio. We followed the basic rule that magnetic pixels with the same pole facing upward cannot be directly connected

to avoid magnetic interference, as the repulsive force will directly separate the two pixels. Figure 6 includes some primitives with desired deformation effects.

We used three magnets (N52, $\varnothing 15$ mm, $H = 5$ mm, some were disassembled from the magnetizer) located beneath the pattern for actuation, ensuring that the north magnetic pole is always facing upwards against the operating surface. In the figures, we have marked the polarity of each magnetic pixel within the *MagPixel* patterns. Pixels with the north facing up are marked in red and labeled as "N" for clear identification. These visual cues will aid the understanding of the orientation and behavior of the magnetic pixels in the design process.

In the explored design primitives (Fig. 6), each pattern displays unique morphing profiles. Generally, primitive patterns with magnetic pixels near the edges typically exhibit localized deformation. In contrast, patterns with magnetic pixels at the center tend to change the overall shape. In summary, by combining these primitive patterns, various deformation effects can be realized, enhancing the design's versatility.

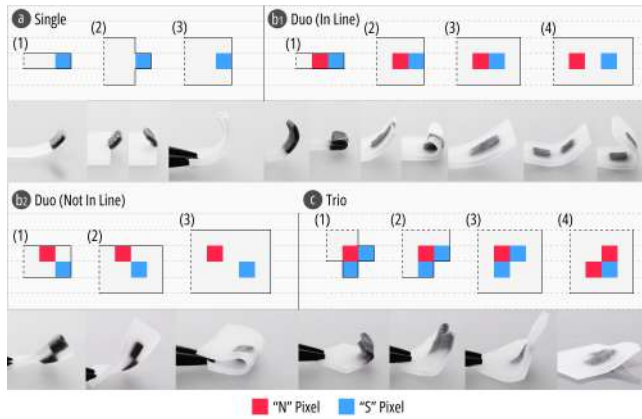


Figure 6: Design primitives and shape-changing effects of *MagPixel* patterns.

4 APPLICATION EXAMPLES

The design and fabrication methods of the *MagPixel* pattern are sufficiently simple and rapid, allowing for basic deformation and movement effects to be achieved through handheld magnets. This accessibility enables individuals without an engineering or technical background to swiftly create prototypes of transformable interfaces. Based on the shape-changing primitives and their corresponding effects we have discussed earlier, we extend our design primitives into more elaborate and interesting applications. For scenarios requiring automated control of the magnetic field or the detection of changes in the magnetic field, the tangible interface composed of *MagPixel* patterns also demonstrates effective interactive capabilities.

4.1 Interactive Storytelling Tools

Utilizing the design structures we have explored, we have simulated various biological forms to construct an interactive storybook (shown in Figure 7) to enhance children's reading experiences.

By manipulating small magnets beneath the pages, the creatures depicted in the storybook respond to the magnets' movements, resulting in various dynamic effects. This interactive approach can help foster children's narrative skills and imagination and familiarize them with the anatomical structures of different creatures. For instance, a vivid emulation can be achieved by moving a permanent magnet up and down directly beneath the four-legged starfish. In another example, rapidly flipping a permanent magnet captures the visual of a slime creature. These examples demonstrate how integrating tangible interactions with educational content can create a more engaging and informative reading experience for children.

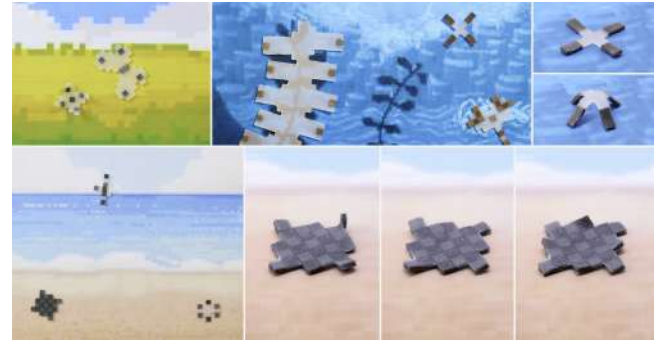


Figure 7: The interactive storybook with moving creatures.

Such design and actuation methods enable the creation of desktop games that do not require any power source. For instance, as shown in Figure 8, we have drawn inspiration from the iconic imagery of the classic video game Pac-Man to design corresponding characters. Players can control their characters by holding permanent magnets and navigating through the game background to accomplish various tasks. This innovative approach offers a unique, tactile gaming experience, merging traditional gameplay into physical interaction.



Figure 8: A tangible version of the video game "Pac-Man".

4.2 Desktop Tangible Interfaces

We have utilized *MagPixel* patterns to create a clock and a reminder for desktops, demonstrating their capabilities as tangible interfaces for display and input purposes. This application showcases the innovative use of *MagPixel* patterns to integrate functional aesthetics with interactive technology, offering users a tangible and intuitive means of engaging with time management and reminders.

To automate the control of the clock example, we use electromagnets as the magnetic source. The electromagnets are connected to a

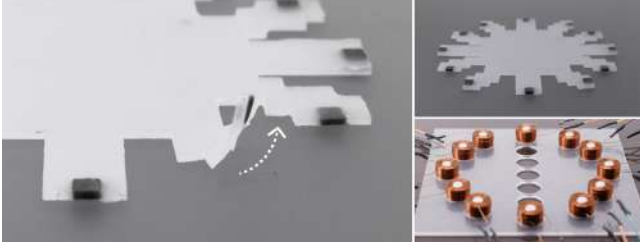


Figure 9: The clock application.

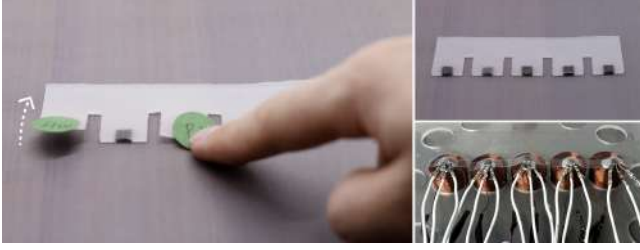


Figure 10: The reminder application.

12V DC power supply, and an Arduino Uno is employed to switch their power state. The Arduino Uno is programmed to send digital high and low digital signals to a relay, which then controls the activation and deactivation of the electromagnets. To effectively manage the considerable heat produced by the electromagnets when energized, we attached heat sinks to their undersides, ensuring sufficient heat dissipation.

For the Reminder example, we placed Hall effect sensors beneath each Reminder to detect the magnetic field changes caused by user presses. When a user completes a specific task and presses down on the Reminder, the Hall effect sensor detects this press event. In response, the Arduino sends a low digital signal to the relay, turning off the corresponding electromagnet and causing the Reminder to drop. Conversely, when a user attaches a new task to the Reminder and presses it, the Hall effect sensor again detects this action. The Arduino then sends a high digital signal to the relay, activating the correct electromagnet, which elevates the Reminder. This system provides an intuitive and interactive way to track task completion.

5 LIMITATIONS AND FUTURE WORK

From our observation, the magnetic strengths of the magnetic pixels are relatively stable, and users can achieve effective and reproducible results, given that the procedure is precisely executed and strictly follows our fabrication guidelines. However, The limitations of the work are as follows:

- (1) For better implementation of *MagPixel*, an in-depth investigation into the magnetic and elastic properties of magnetic pixels is needed. For example, the magnetic hysteresis. After these investigations, we can readjust the ratio of the ingredients and the dimension of the magnetic pixel to optimize the shape-changing performance. Additionally, we used off-the-shelf NdFeB magnets for magnetization, and the magnetic field strength at the center point of the top magnet can reach

approximately 790 mT, which is not enough to saturate the magnetic pixel. The external magnetic field, more specifically, the actuation apparatus that controls and drives the shape-changing display, also warrants future discussion.

- (2) The magnetization process charges the top and bottom surfaces of magnetic pixels with opposite polarities. Due to the demagnetizing field, the surface exhibits an uneven distribution of magnetic fields, which may negatively impact the shape-changing performance.
- (3) During the curing process of the magnetic pixels, the magnetic particles can form sedimentation at the bottom of the mixture, resulting in an uneven concentration within the magnetic pixels, undermining the uniformity of the magnetic strengths after magnetization.
- (4) Strong external magnetic field opposite to the magnetic pixel may result in its irreversible demagnetization. This phenomenon is observed and discussed in existing literature [22].
- (5) In this paper, the *dynamic characteristic* of the magnetic pixel has not been discussed. Due to the flexibility of the material, the magnetic self-interaction inside the pixel will cause real-time changes in the internal stresses during deformation, such as bending, curling, and folding. Thus, the energy equation and the kinetic equation may have strong non-linearity.

For future work, we plan to provide users with an interactive design tool that can quickly preview the shape-changing effects, with visual cues such as color coding to indicate the magnetic poles and other features such as detailed instructions for the magnetization procedures (e.g., magnetize the pixels in other directions). To get better shape-changing effects, we plan to provide a more comprehensive actuation toolkit reconfigured from the magnetizer. We will also discover the possibility of magnetic voxels made from the same magnetic soft materials, adding another dimension to the design space of *MagPixel*.

6 CONCLUSION

In this paper, we present *MagPixel*, a cost-efficient and user-friendly toolkit that democratizes the design and fabrication of interactive shape displays using magnetic soft materials. Traditionally limited to specialized fields of material science and robotics, the utilization of these materials in Human-Computer Interaction (HCI) is expanded by *MagPixel*. The toolkit simplifies the complex fabrication process into accessible, standardized procedures with a modular toolkit. The grid-based design method of *MagPixel* ensures scalability and flexibility, allowing even novices and DIY enthusiasts to create responsive, shape-changing interfaces. This innovation fosters creativity within HCI and invites participation outside of HCI, paving the way for diverse applications of magnetic soft materials for a broader audience.

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